

Object-Conditioned Cross-Channel Mechatronic Parameter Reconfiguration for Haptic Teleoperation of Heterogeneous Objects

Fengjie Ma^a, Chuang Cao^b, Hua Zhang^{a,*}, Qixin Cao^b, Yilin Zhou^a

^a*School of Mechanical and Automotive Engineering, Shanghai University of Engineering Science, 333 Longteng Road, Shanghai, 201620, China*

^b*School of Mechanical Engineering, Shanghai Jiao Tong University, 800 Dongchuan Road, Shanghai, 200240, China*

Abstract

Fixed teleoperation settings cannot readily accommodate objects with different fragility, geometry, and grasping requirements. Existing object-conditioned approaches primarily adapt slave-side impedance, while coordinated configuration of the slave response, operator-facing haptic interface, and gripper execution remains less explored. This paper presents an object-conditioned framework that maps object semantics to a common operational strategy and jointly configures these three mechatronic subsystems. Low-rate visual perception is integrated with a nominal 200-Hz supervisory loop through asynchronous, non-blocking communication, and the first valid detection triggers a one-shot parameter update that remains fixed for the trial. Three operators completed 135 grasp-and-transfer trials involving six objects and five control modes. Compared with impedance-only scheduling, the fully coordinated mode reduced mean completion time by 1.79 s, corresponding to 8.5%, with a matched-block bootstrap 95% confidence interval of [1.10, 2.51] s. The direction of the difference was consistent across all operators and objects, while the coordinated mode also showed the highest observed success rate and the lowest Raw NASA-TLX score. These results provide system-level evidence that jointly configuring the haptic interface and gripper with slave-side impedance can improve heterogeneous-object teleoperation under the present experimental conditions.

Keywords: haptic teleoperation, impedance control, telemanipulation, object-conditioned parameter reconfiguration, cross-channel parameter coordination, human-in-the-loop evaluation

1. Introduction

Teleoperation leverages human judgment to perform remote manipulation tasks that remain difficult for autonomous robots, particularly in hazardous environments, flexible manufacturing, and unstructured settings. Haptic feedback conveys remote contact information to the operator and supports assessment of contact state, grasp stability, and slip. Previous studies have examined the stability and transparency of bilateral teleoperation and the adaptation of controllers to the environment, operator, and task [1, 2]. In practical systems, however, these control functions must operate in concert with visual perception, mechanical execution, and the human-machine interface.

For heterogeneous objects, a single parameter set must balance gentle contact with positioning and transfer stability. Lower stiffness and gripping force can reduce contact severity but may compromise positioning and transfer stability, whereas higher values may improve manipulation speed at the cost of greater impact or deformation risk. The six test objects—an apple, a banana, a paper cup, a bottle, a computer mouse, and a pair of scissors—were selected to represent diverse demands on compliance, haptic interaction, and gripper execution. This controlled object set enabled evaluation of coordinated parameter reconfiguration for heterogeneous grasping tasks.

Impedance control regulates the relationship among displacement, velocity, and interaction force, allowing a robot to remain compliant during contact [3]. Fixed impedance is straightforward to implement but inevitably represents a compromise across objects. Variable-impedance methods instead adapt stiffness and damping according to task state, contact information, or learned policies, while accounting for the stability implications of time-varying parameters [4, 5]. In teleoperation, operators may adjust slave-side impedance directly [6], whereas tele-impedance approaches map human motion and impedance information to the remote robot [7–9]. Such continuous adaptation commonly relies on human measurements, demonstration data, or post-contact feedback.

Visual information has also been used to select or adapt robot impedance. A vision-and-voice tele-impedance method selected slave-side impedance from object properties while retaining an operator confirmation or correction step [10], and visual-impedance control has combined visual and force information in a visual-feature space [11]. At the workflow level, intention inference and task-oriented shared control have reduced manual decision-making [12]. More recent approaches have derived stiffness from object geometry, material, and environmental relationships [13] or generated task-relevant stiffness matrices from visual, gaze, and language inputs [14]. These studies primarily translate task information into slave-side stiffness, impedance, or control-authority decisions.

Automatic switching and passive arbitration additionally de-

*Corresponding author

Email address: huazhang@yeah.net (Hua Zhang)

termine how control authority is allocated and when it changes [15, 16]. Haptic interfaces have been evaluated for telemanipulation performance [17], while related work has addressed compliant assembly [18], human and environmental stiffness estimation [19], rotational impedance [20], scaling transitions [21], vision-based tactile feedback [22], multi-stiffness interfaces [23], delay-aware shared autonomy [24], and haptic support under restricted vision [25]. These studies generally adapt individual control or interaction elements. By contrast, limited work has used object semantics as a common task condition to jointly configure the slave-side mechanical response, the operator-facing haptic interface, and gripper execution. The central question is therefore not whether vision can tune impedance, but whether a strategy-level semantic decision can configure these three mechatronic subsystems as a unified parameter bundle and provide task-level benefits beyond visual-semantic display, manual selection, and impedance-only scheduling.

To address this gap, we propose a vision-semantic, event-triggered multi-channel joint parameter-configuration framework for pre-contact parameter preparation in haptic teleoperation of heterogeneous objects. A three-level mapping from object semantics to an operational strategy and then to a joint parameter vector translates task information into coordinated settings for slave-side impedance, the master-side haptic interface, and gripper execution. After task initiation, the first detection above the confidence threshold with a valid strategy mapping triggers a one-time update to the target parameters. The selected strategy is then locked for the remainder of the trial to prevent repeated switching in response to subsequent detection fluctuations.

The central positioning of this work is a vision-semantic, event-triggered multi-channel joint parameter-configuration framework for pre-contact parameter preparation in haptic teleoperation of heterogeneous objects. The framework uses object semantics as a task prior and aims to make the object-conditioned parameter bundle available before interaction, rather than continuously adapting it after contact. Its online stage uses discrete, event-triggered updates rather than post-contact continuous optimization or persistent visual servoing. Asynchronous implementation and strategy locking keep low-rate perception outside the nominal supervisory control path; by avoiding complex online optimization, the framework enables interpretable and deployable system-level parameter configuration with a relatively low online computational burden.

The system-level evaluation uses five modes: fixed parameters, operator selection, full cross-channel reconfiguration, a visual-semantic cue with fixed parameters, and impedance-only scheduling. The C–D comparison assesses automatic reconfiguration beyond the displayed visual-semantic cue. The primary C–E comparison holds the visual-semantic mapping and slave-side impedance settings constant and tests the complete parameter bundle against impedance-only scheduling. Operator- and object-stratified results describe whether this system-level difference has a consistent direction. They do not identify the independent causal contributions of the haptic and gripper parameter groups.

The main contributions are as follows:

1. **Event-triggered multi-channel joint parameter configuration for pre-contact preparation.** The framework extends object-dependent configuration from slave-side impedance alone to three jointly configured subsystems: the slave-side mechanical response, the operator-facing haptic interface, and gripper execution. A common operational strategy assigns a consistent parameter bundle; the joint configuration is not derived from a dynamic coupling model between channels.
2. **Three-level semantic mapping within a constrained parameter space.** Object classes are grouped into fragility-priority, balanced, and stability-priority operational strategies and mapped to a seven-dimensional joint parameter vector constrained by hardware capabilities, task risks, and the operating envelope established in preliminary testing.
3. **Asynchronous, event-triggered one-shot reconfiguration.** Image acquisition at 15 fps and independent YOLO11n inference communicate with a nominal 200-Hz supervisory update through bounded queues. Non-blocking polling prevents the supervisory thread from waiting for perception; strategy locking avoids repeated intra-trial switching caused by detection fluctuations; and an approximately 300-ms transition smooths changes in stiffness and the controller damping-scaling parameter.
4. **Human-in-the-loop system-level evaluation across five modes.** Five modes were implemented on an Omega.7–Panda platform to provide comparative evidence for visual-semantic display, manual strategy selection, impedance-only scheduling, and full cross-channel reconfiguration. The primary C–E comparison evaluates the complete parameter bundle at the system level and does not separate the individual effects of the added haptic-interface and gripper channels. The evaluation involved three operators, six objects, and 135 trials.

2. Methods and System Implementation

2.1. Mechatronic System Architecture and Asynchronous Execution

The experimental platform comprised an Omega.7 force-feedback master device, a seven-degree-of-freedom Franka Panda robot, a Franka Hand gripper [26], an Intel RealSense D435i camera, and a control computer (Fig. 1). The D435i provided 424×240 -pixel color images at 15 fps; the depth stream was not used. The control computer ran visual detection, the nominal 200-Hz supervisory teleoperation update, parameter configuration, haptic rendering, gripper control, and data logging.

Figure 2 illustrates the timing relationships within this multi-rate asynchronous software architecture.

RGB frames are acquired every 66.7 ms and written to a single-slot queue, preventing stale images from accumulating. YOLO11n runs in an independent process and returns class labels and confidence scores through a two-slot result queue. The nominal 200-Hz supervisory loop polls this queue every 5 ms

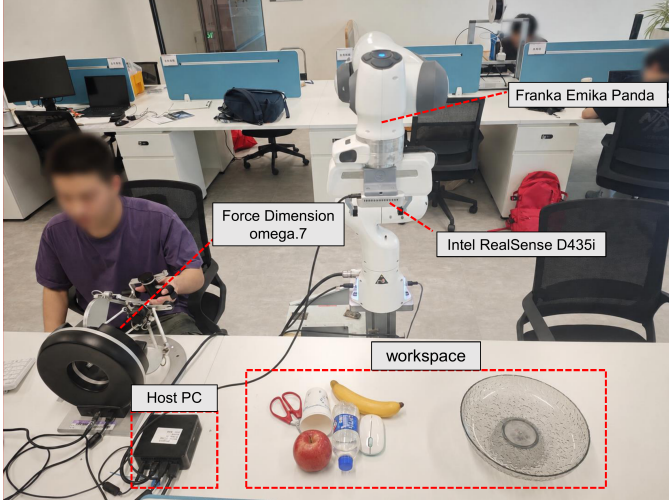


Figure 1: Experimental platform comprising the Omega.7 master device, Franka Panda robot and Franka Hand gripper, D435i camera, and control computer. The camera supplied 424×240 color images at 15 fps.

without waiting for inference, allowing camera acquisition, perception, and operator motion to proceed in parallel. The reported 48.19 ms is the mean wall-clock processing time per image rather than the camera sampling period.

2.2. Master-Slave Mapping and Implementation of the Mechatronic Channels

Only translational motion of the Omega.7 was mapped to the Panda. The master-device position increment and desired slave position were updated as

$$\Delta \mathbf{p}_m(k) = \mathbf{p}_m(k) - \mathbf{p}_m(k-1), \quad (1)$$

$$\mathbf{p}_d(k) = \mathbf{p}_d(k-1) + S \mathbf{C} \Delta \mathbf{p}_m(k), \quad (2)$$

where the position-scaling factor was $S = 3.0$ and $\mathbf{C} = \text{diag}(-1, -1, 1)$. The desired end-effector orientation $\mathbf{R}_d$ was fixed at its prescribed value throughout each trial; active rotational input from the operator was not mapped. Cartesian impedance control acted on the six-dimensional pose error,

$$\mathbf{w}_c = \mathbf{K}(c) \mathbf{e} + \mathbf{D}(c) \dot{\mathbf{e}}, \quad (3)$$

$$\mathbf{K}(c) = \text{diag}(K_t, K_t, K_t, K_r, K_r, K_r), \quad (4)$$

where $\mathbf{e}$ contains translational error relative to $\mathbf{p}_d$ and rotational error relative to the fixed $\mathbf{R}_d$. Thus, K_r changes resistance to rotational disturbances and the ability to hold the prescribed orientation; it does not scale operator-commanded rotation.

The slave controller internally generated the damping matrix from the active diagonal stiffness matrix and a scalar API parameter denoted by ζ . In the present implementation,

$$\mathbf{D}(c) = 2\zeta(c) \sqrt{\mathbf{K}(c)}, \quad (5)$$

where the square root is applied elementwise to the diagonal entries. Here, ζ is the dimensionless damping-scaling parameter used by the controller. We do not interpret it as a Cartesian critical-damping ratio because no effective Cartesian mass or inertia matrix was identified or specified.

For haptic rendering, the Franka internal state estimator provided the slave-side external-force estimate $\mathbf{F}_{\text{ext}}$ in newtons. The baseline master-side force command along direction i was

$$u_{h,i}^{\text{base}} = \text{sgn}(K_f F_{\text{ext},i}) \max(|K_f F_{\text{ext},i}| - d, 0), \quad i \in \{x, y, z\}, \quad (6)$$

where K_f is a dimensionless force-scaling factor and d is a per-axis dead-zone threshold in newtons.

A separate positive- z cue was generated solely from the Omega.7 gripper angle θ_Ω :

$$\alpha_g = \text{clip}\left(\frac{\theta_{\max} - \theta_\Omega}{\theta_{\max} - \theta_{\min}}, 0, 1\right), \quad (7)$$

$$u_g = \min(\alpha_g G_{\text{cue}} F_{\text{cue,max}}, F_{\text{cue,max}}), \quad u_{h,z} = u_{h,z}^{\text{base}} + u_g. \quad (8)$$

With $G_{\text{cue}} = 0.3$ and $F_{\text{cue,max}} = 1.0$ N, the cue was $u_g = 0.3\alpha_g$ N and therefore remained within $[0, 0.3]$ N. It was updated during every nominal 200-Hz supervisory cycle, with a more open Omega.7 gripper producing a larger cue. This signal represents only the Omega.7 gripper aperture; it contains no measurement of Franka Hand aperture, master-slave aperture error, grasp force, or button state.

For gripper execution, v_g was passed as the speed argument of the grasp API, whose remaining arguments were width, force, epsilon_inner, and epsilon_outer; it was also used by move during opening and non-grasp position adjustments. The strategy value F_g specified the requested force argument of grasp() in newtons; it was not an independently measured fingertip force. The target grasp width was 0 m. After contact and successful entry into the HOLDING state, the grasp command was not repeatedly retransmitted; the gripper maintained the grasp internally until stop() and the subsequent release or opening command.

Together, the three parameter channels act on distinct elements of the teleoperation system. Panda impedance defines the slave-side response to pose error and disturbance, the Omega.7 settings determine how the estimated external force is rendered to the operator, and the Franka Hand settings govern grasp execution. The proposed method assigns these settings in parallel from a single operational strategy. Coordination therefore occurs at the strategy layer through a consistent parameter bundle; no dynamic coupling constraint or master-slave impedance-matching law is derived between channels. Haptic transparency and closed-loop fingertip-force accuracy were not evaluated.

2.3. Object-Conditioned Cross-Channel Mechatronic Parameter Reconfiguration

The framework uses a three-level mapping from detected object class to an operational strategy and then to a joint parameter vector,

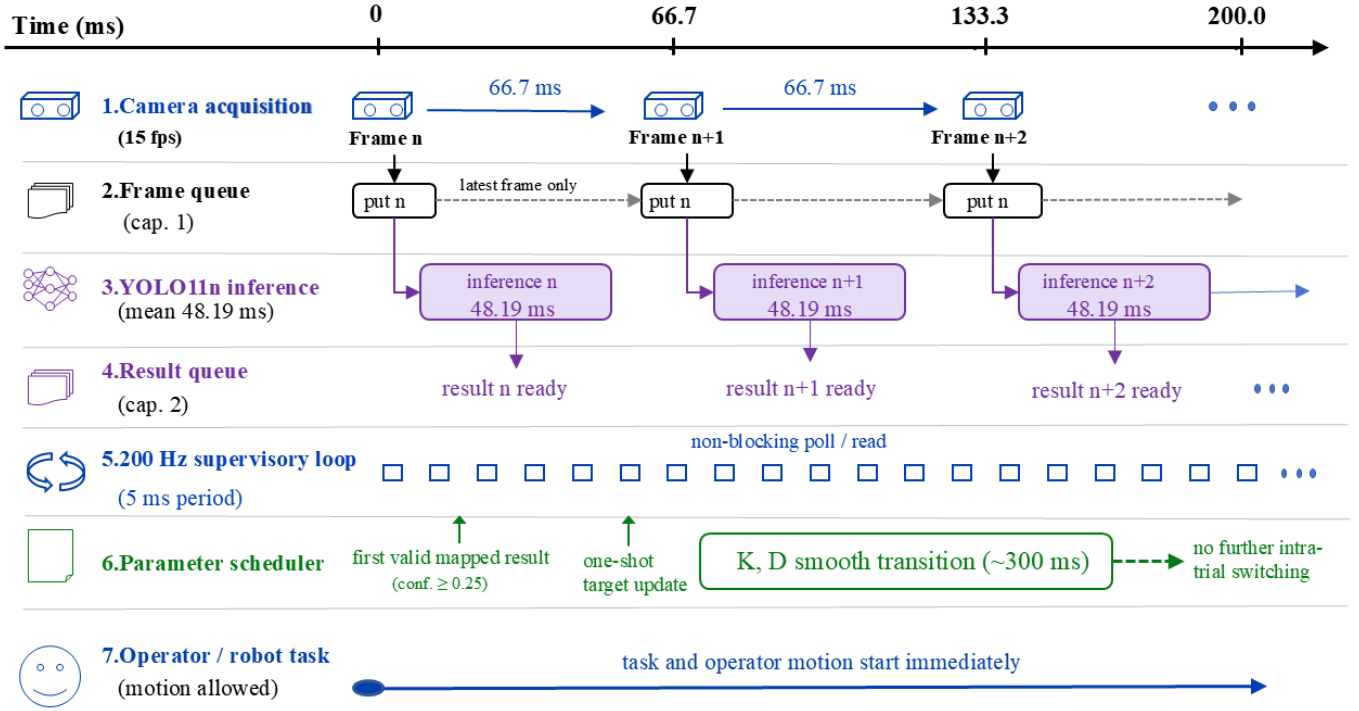


Figure 2: Illustrative timing of the asynchronous perception-control implementation. Color frames are acquired at 15 fps and passed through a single-slot frame queue to an independent YOLO11n process; a controlled visual test yielded a mean wall-clock processing time of 48.19 ms per image. Class labels and confidence scores are returned through a two-slot result queue. The nominal 5-ms supervisory update polls this queue without blocking. The first result that reaches the confidence threshold is converted by the class-to-strategy mapping into a one-shot strategy event. Modes C and E update their enabled parameters, whereas mode D retains fixed parameters and presents the semantic result. Translational stiffness, rotational stiffness, and the controller damping-scaling parameter follow an approximately 300-ms smoothstep transition, while the enabled haptic-interface and gripper settings change at the event. Subsequent detections do not change the strategy within the same trial. A low-confidence result or absence of a detection leaves the initialized state unchanged; a detected class without an explicit mapping is assigned to the balanced default strategy.

Table 1: Runtime characteristics and data flow of the multi-rate mechatronic system.

Module	Timing or trigger	Data flow
Master input	200 Hz	Omega.7 translational position and buttons → master-state update.
Slave control	200 Hz	Desired pose and impedance parameters → Panda command.
Image acquisition	15 fps (66.7 ms/frame)	D435i color stream → 424 × 240 color image.
YOLO11n inference	Mean 48.19 ms/image; asynchronous	Latest available image → class and confidence.
Strategy scheduling	First threshold-qualified detection	Class and confidence → target $\Theta(c)$.
Haptic rendering	200 Hz	Force-related state → Omega.7 force command.
Gripper command	Button event	Gripper input and command parameters → Franka Hand command.

Note: All modules operate non-blockingly with respect to the supervisory loop. YOLO11n inference is executed in an independent process.

$$\Theta(c) = \{K_t, K_r, \zeta, K_f, d, v_g, F_g\}. \quad (9)$$

The strategy classes represent combined grasping risks rather than material softness alone. The apple and banana were assigned to the fragility-priority strategy because lower contact impact, closing speed, and requested grasp force were preferred to limit squeezing and deformation, while their smooth surfaces still posed a risk of slip. The paper cup and bottle were assigned to the balanced strategy. Excessive force can deform a

paper cup, whereas overly low settings may produce an insecure grasp; the bottle similarly requires a compromise between gentle handling and slip resistance. The mouse and scissors were assigned to the stability-priority strategy because their smooth or irregular surfaces and transfer demands—and, for the scissors, elongated geometry—placed greater emphasis on pose retention and grasp stability. These assignments were specific to the present objects, platform, and task.

At run time, the first threshold-qualified result read by the

Table 2: Parameter settings for the three operational strategies.

Strategy	Assigned objects	Slave impedance	Haptic interface	Gripper
Fragility-priority	Apple, banana	$K_t = 50 \text{ N/m}$; $K_r = 5 \text{ N m/rad}$; $\zeta = 0.8$	$K_f = 0.2$; $d = 0.3 \text{ N}$	$v_g = 0.02 \text{ m/s}$; $F_g = 8 \text{ N}$
Balanced	Paper cup, bottle	$K_t = 150 \text{ N/m}$; $K_r = 10 \text{ N m/rad}$; $\zeta = 1.0$	$K_f = 0.5$; $d = 0.4 \text{ N}$	$v_g = 0.05 \text{ m/s}$; $F_g = 15 \text{ N}$
Stability-priority	Mouse, scissors	$K_t = 200 \text{ N/m}$; $K_r = 13 \text{ N m/rad}$; $\zeta = 1.2$	$K_f = 0.7$; $d = 0.5 \text{ N}$	$v_g = 0.10 \text{ m/s}$; $F_g = 20 \text{ N}$

supervisory loop triggered a one-time update of the parameters enabled in the active mode, after which the strategy remained locked for the rest of the trial. Translational stiffness, rotational stiffness, and the damping-scaling parameter followed a smoothstep interpolation over approximately $T_s = 0.30 \text{ s}$. With $\tau = \text{clip}((t - t_0)/T_s, 0, 1)$ and $s(\tau) = 3\tau^2 - 2\tau^3$, each interpolated quantity $q \in \{K_t, K_r, \zeta\}$ was updated as

$$q(t) = q_0 + s(\tau)(q_{\text{target}} - q_0). \quad (10)$$

The controller recomputed $\mathbf{D}$ from the current $\mathbf{K}$ and ζ during this transition. Enabled values of K_f , d , v_g , and F_g changed immediately at the strategy event. Because operator motion was not gated by perception, the mechanism is described as a one-shot update after task initiation. The available logs do not allow its timing relative to first contact to be reconstructed for individual trials.

The class-to-strategy mapping is manually engineered. If an existing detector already provides a stable label for a new object, that label can be added to the mapping without retraining the controller. If the detector cannot identify or distinguish the object reliably, it must be retrained or replaced before a strategy can be assigned. The present system does not automatically infer fragility, friction, or other physical attributes for an unseen class.

2.4. Experimental Modes and Parameter Rationale

The fixed baseline vector was

$$\Theta_0 = (150, 10, 1.0, 0.5, 0.3, 0.10, 20), \quad (11)$$

with the same element order as $\Theta(c)$. Table 3 summarizes the five experimental conditions and the comparisons they support.

The camera window could be closed by pressing q . Table 3 describes the default software behavior; the window state was not included in the formal trial logs.

The three strategies were represented by discrete operating points. Before the formal experiment, two researchers screened candidate combinations on representative objects and excluded settings associated with sustained oscillation, clear haptic discomfort, grasp failure, or visible object damage. The final values were fixed before the 135 formal trials. The fragility-priority point used 50 N/m translational stiffness, 8 N requested grasp force, and 0.02 m/s gripper speed to reduce contact and gripper-closing severity. The balanced point used 150 N/m, 15 N, and

0.05 m/s to avoid both excessive deformation and unreliable holding. The stability-priority point used 200 N/m, 20 N, and 0.10 m/s to favor pose retention and transfer stability. These operating points were selected qualitatively for the present setup and should not be interpreted as material-specific damage limits.

2.5. Execution Procedure

Algorithm 1: Object-conditioned cross-channel parameter scheduling

1. Load the mode-specific initialization in Table 3 and start the nominal 200-Hz supervisory update; launch the visual process only in modes C–E.
2. Run object detection asynchronously. The supervisory thread polls the results without blocking and checks the confidence threshold.
3. If no detection meets the threshold, retain the initialized state. If the detected label has no explicit class-to-strategy entry, assign the balanced default strategy.
4. Use the first threshold-qualified strategy to update the parameter channels enabled by the current mode. Interpolate K_t , K_r , and the controller parameter ζ over approximately 300 ms and update the enabled haptic-interface and gripper settings immediately.
5. Lock the selected strategy for the remainder of the trial and continue teleoperation, haptic rendering, and gripper execution without further visual switching.
6. Record the active parameter state, task trajectory, duration, and outcome, and reset the system after the prescribed terminal procedure.

2.6. Bounded Fallback and Safety Measures

A misclassification as another known class invoked the strategy assigned to that class. A detected label without an explicit mapping invoked the balanced default strategy, whereas a missing or low-confidence detection caused no update. Every fallback outcome was confined to one of the three predefined parameter sets, although a bounded parameter set may still be inappropriate for a misclassified object. Per-trial visual events were outside the formal logging scope, so misclassifications and default-branch activations were not individually recorded.

In addition to these software fallbacks, robot collision detection, Franka Hand force settings, zero-force commands on exit, a standardized initial pose, and a manual emergency stop

Table 3: Experimental conditions, channel-update logic, and comparison roles for the five modes.

Mode	Visual cue / selection	Slave impedance	Haptic interface	Gripper execution	Comparison role
A: Fixed parameters	None	—	—	—	Fixed-control baseline.
B: Operator selection	No visual cue; operator selects a complete strategy after timing starts.	Manual	Manual	Manual	Manual-selection workflow baseline.
C: Full cross-channel	Class, strategy, confidence, and colored bounding box	Auto	Auto	Auto	Proposed method; C–D tests automatic reconfiguration beyond the visual-semantic cue, and C–E tests the added haptic and gripper channels.
D: Visual-semantic cue only	Same as C	—	—	—	Isolates the displayed visual-semantic cue.
E: Impedance only	Same as C	Auto	—	—	Isolates impedance-only reconfiguration.

Note: Auto denotes a one-shot update after the first valid visual detection; Manual denotes operator selection after timing starts; — retains the initialized setting. In modes C–E, no detection or confidence below 0.25 retains the initialized state. A detected class without an explicit mapping invokes the balanced strategy; in mode D, only its label is displayed and Θ_0 is retained. Vision is not used in modes A and B.

Table 4: Parameter design space, engineering constraints, and selected levels.

Parameter	Low-to-high operational effect	Engineering constraint	Selected levels
Slave impedance: K_t	Greater translational compliance and lower impact $\rightarrow$ greater positional stability.	Stable Panda impedance-control range.	50 / 150 / 200 N/m
Slave impedance: K_r	Greater rotational compliance about the fixed desired orientation $\rightarrow$ greater resistance to rotational disturbance.	Stable rotational response.	5 / 10 / 13 N m/rad
Slave impedance: ζ	Lower $\rightarrow$ higher controller damping scaling.	Avoid sustained oscillation and excessive sluggishness.	0.8 / 1.0 / 1.2
Haptic interface: K_f	Weaker $\rightarrow$ stronger rendered external-force cue.	Acceptable Omega.7 response.	0.2 / 0.5 / 0.7
Haptic interface: d	Greater sensitivity to small forces $\rightarrow$ greater suppression of small signals.	Haptic-interface jitter and noise.	0.3 / 0.4 / 0.5 N
Gripper: v_g	Lower-impact motion $\rightarrow$ faster grasp establishment.	Franka Hand execution range.	0.02 / 0.05 / 0.10 m/s
Gripper: F_g	Lower $\rightarrow$ higher requested grasp force.	Franka Hand force setting.	8 / 15 / 20 N

provided operational safeguards. The Omega.7 was operated using its standard driver and hardware configuration. The implemented mapping limited the additional master-gripper-aperture cue to 0.3 N. The baseline force-feedback command was transmitted without an additional software saturation layer, and saturation status was not logged.

3. Experimental Design

3.1. Evaluation Objectives and Analysis Plan

The primary evaluation objective was to quantify task completion time under the complete cross-channel configuration relative to the planned baselines. Success rate and Raw NASA-TLX were reported as supplementary descriptive evidence of task success and subjective workload. Master-side trajectory length and pause count were included as exploratory process measures. Mode A served as the fixed-control baseline; mode B represented a workflow baseline that included manual object identification and strategy selection after timing began; and mode D represented the visual-semantic-cue baseline. Mode E retained vision-semantic impedance scheduling but not the haptic-interface or gripper settings. The primary C–E comparison therefore evaluated the complete parameter bundle at the

system level against impedance-only scheduling; it was not designed to separate the independent contributions of the added parameter groups.

3.2. Operators and Test Objects

Three operators completed the main experiment (P01–P03; 23–24 years old; two male and one female; all right-handed). The Omega.7 was positioned to the operator’s left and was operated with the left hand in every trial. All operators had received basic teleoperation training and completed an additional 10–15 min familiarization session before each formal experiment to practice master-device motion, gripper input, and the task procedure. All participants provided written informed consent.

The six objects were selected to represent different contact, grasping, and transfer requirements and were assigned to the fragility-priority, balanced, or stability-priority strategy. These assignments were used solely to determine the parameter settings in this experiment and should not be interpreted as a general classification of material properties.

3.3. Experimental Modes and Trial Structure

The five modes and their comparison logic are defined in Table 3. Modes C, D, and E used the same default visual-semantic display. The C–D contrast therefore compares full au-

Table 5: Test objects, assigned strategies, and principal task risks.

Object	Assigned strategy	Physical profile	Principal task risk
Apple	Fragility-priority	~200 g; smooth; $\varnothing$ 70–80 mm	Impact or slip; gentle contact required.
Banana	Fragility-priority	~120 g; smooth; 20 × 180 mm	Compression-induced deformation and slip.
Paper cup	Balanced	~5 g; paper; $\varnothing$ 75×90 mm	Easily deformed; stable grasp required.
Bottle	Balanced	~30 g; smooth plastic; $\varnothing$ 65×200 mm	Slip; balance between efficiency and stability.
Mouse	Stability-priority	~100 g; smooth plastic; 65 × 120 × 35 mm	Rigid irregular surface; transfer slip.
Scissors	Stability-priority	~150 g; metal and plastic; 50 × 170 × 15 mm	Rigid elongated geometry; high orientation demand.

Table 6: Allocation of matched task blocks and trials by strategy and object.

Object	Blocks	Trials
<i>Fragility-priority</i>		
Apple	4	20
Banana	5	25
<i>Balanced</i>		
Paper cup	5	25
Bottle	4	20
<i>Stability-priority</i>		
Mouse	5	25
Scissors	4	20
Total (six objects)	27	135

Note: Each block contains one trial in each of the five modes.

tomatic parameter reconfiguration with displayed semantic information alone, whereas C–E compares full cross-channel reconfiguration with impedance-only reconfiguration while holding the designed interface constant. The D–A contrast assesses the complete visual-semantic cue—class, mapped strategy, confidence, and bounding box—and does not isolate the class label. Mode B required manual object identification and button-based strategy selection after timing had started but before robot motion could begin. Its comparison with mode C therefore includes both automation of the selection workflow and differences in visual-interface presentation; the separate decision time was not logged.

The experiment comprised 27 matched task blocks. Each block was defined by a fixed operator, object, and repetition index and contained one trial in each of modes A–E, yielding $27 \times 5 = 135$ trials. Within each operator and strategy category, the two objects were assigned to the three repetitions in a fixed 2:1 ratio. This assignment was held constant across the five modes for a given operator and strategy, preserving the matched comparison, while the 2:1 allocation was reversed across operators. The resulting object counts produced an approximate 5:4 balance within each strategy:

The five mode orders were preassigned at the operator-by-strategy group level using nine different sequences, so no single fixed order was used. The schedule distributed modes across early and late positions but did not constitute a complete Latin-square design. The exact sequences are reported in Supplementary Table S2. With only three operators, order, object, and

operator effects could not be fully separated.

3.4. Task and Procedure

Before each trial, the test object was manually placed within a predefined work region and at an approximately prescribed orientation. Small reset errors in position and orientation were therefore unavoidable, but the same placement protocol was used in all modes. Only translational master motion was mapped; the desired Panda end-effector orientation remained fixed throughout the trial.

Each trial comprised reset, approach, grasp, transfer, release, and termination (Fig. 3). In modes C–E, camera acquisition, visual inference, and teleoperation began concurrently at task initiation; visual output did not gate operator motion. In mode B, timing began before object identification and strategy selection, and robot motion was permitted only after the operator completed the button selection. No separate timestamp marked the end of this selection step.

A trial was classified as successful when grasping, transfer, and placement were completed without a drop, observable slip, or visible damage. Complete separation of the object from the gripper was classified as a drop; visible relative motion between object and gripper during grasp or transfer was classified as slip; and a clear indentation, crushing deformation, or other visible damage was classified as damage. A designated researcher applied these predefined criteria in real time. Trials were not systematically video-recorded, and the labels were not independently or blindly re-evaluated.

To maintain a common timing endpoint, failure did not stop the timer immediately. Regrasping after a drop was not permitted; the failure was recorded, and the operator completed the prescribed return-and-termination procedure until the robot and gripper reached the predefined terminal state. The same endpoint was therefore applied to successful and failed trials. The system recorded the master-side trajectory, gripper input, active control parameters, task duration, and outcome.

3.5. Evaluation Measures

Task completion time was the primary endpoint. Success rate was reported descriptively, and master-side trajectory length and pause count were used as exploratory process measures. Subjective workload was assessed with Raw NASA-TLX [27]. Each of the six dimensions was rated from 0 to 100, and their

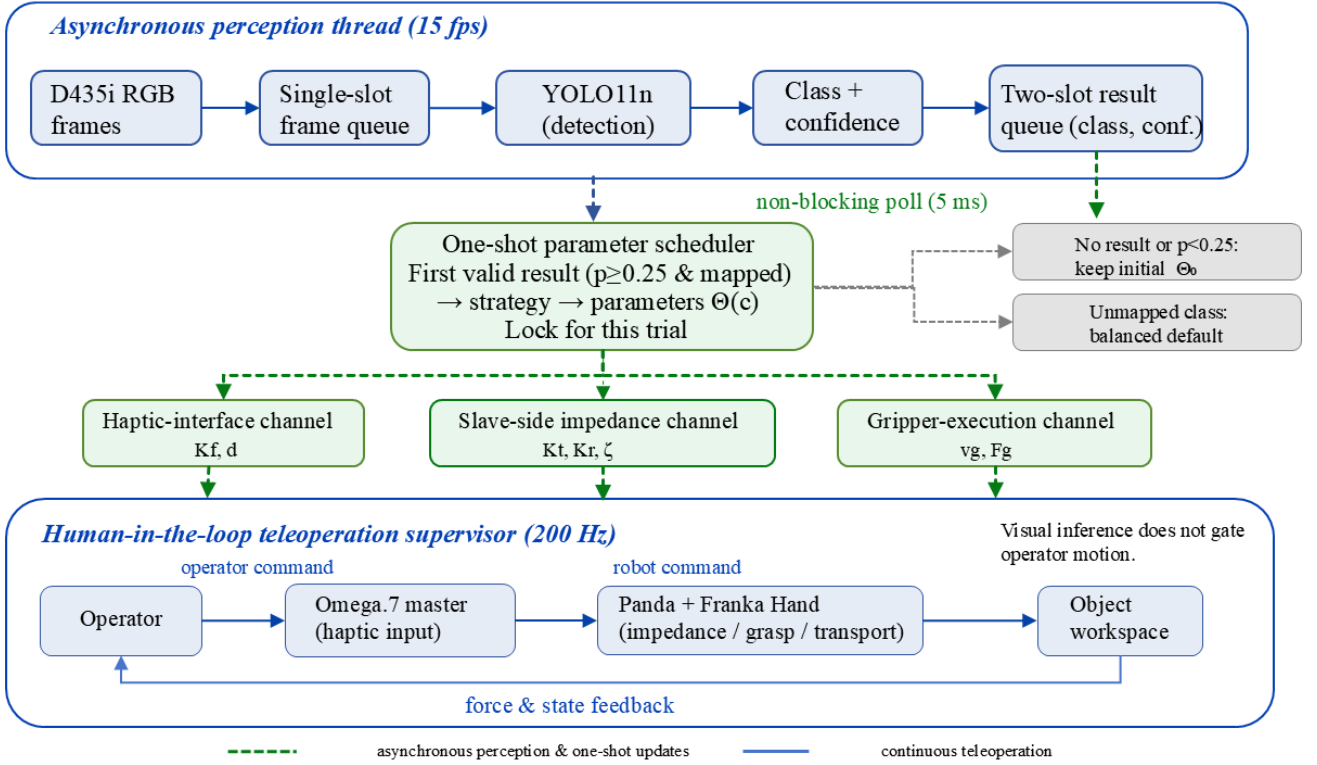


Figure 3: Object-conditioned cross-channel reconfiguration architecture. Color frames acquired at 15 fps are processed asynchronously; the nominal 200-Hz supervisor polls the bounded latest-result queue without waiting for inference. The first threshold-qualified mapped class triggers one strategy event $\Theta(c)$, which updates the enabled slave-impedance, haptic-interface, and gripper settings and is then locked for the trial. No result or confidence below 0.25 retains the initial state, whereas a detected but unmapped class invokes the balanced default strategy. Green paths show asynchronous perception and one-shot updates; the blue loop shows continuous human teleoperation and force/state feedback. Five-mode channel-update logic and comparison roles are given in Table 3.

unweighted arithmetic mean was used as the Raw NASA-TLX score.

Each operator completed one questionnaire immediately after the three trials in a given strategy $\times$ mode condition. These trials combined the two objects assigned to that strategy in the operator-specific 2:1 ratio. The workload dataset therefore contained 3 operators $\times$ 3 strategies $\times$ 5 modes = 45 questionnaires. The resulting scores represent strategy-level workload under each mode and do not support object-specific workload comparisons.

Visual performance was characterized in a separate controlled image test using class accuracy, class-to-strategy mapping accuracy, confidence, and per-image processing time. A pause was defined as a continuous interval during which master-device speed remained below 0.005 m/s for at least 0.30 s. Pauses were detected by differentiating the raw master-side trajectories, which were sampled at approximately 200 Hz.

3.6. Statistical Analysis

Completion time across the five modes was first compared using a Friedman test, with the task block—defined by the same operator, object, and repetition index—as the pairing unit. When the omnibus test was significant, paired Wilcoxon signed-rank tests were performed with Holm–Bonferroni correction for multiple comparisons. For the primary C–E comparison, we report the paired mean difference, relative change, and a 95%

bootstrap confidence interval based on 10,000 resamples of the matched task blocks. We also examined effect direction for each operator and performed a leave-one-operator-out analysis to assess whether a single operator dominated the overall difference. Raw NASA-TLX scores were paired descriptively across nine operator $\times$ strategy units per mode; each score summarized three mixed-object trials, and no bootstrap interval was calculated. Trajectory length and pause count were treated as exploratory measures, whereas success rate was reported descriptively. Because the 27 task blocks were nested within only three operators, the inferential tests characterize repeated-measures differences within the present sample and do not support population-level inference across operators. Results are reported as both median [IQR] and mean $\pm$ SD. Figures 4–9 were generated from frozen data using verified Python/Matplotlib scripts.

4. Results

4.1. Visual Recognition, Semantic Mapping, and Asynchronous Integration

The separate visual test set contained 30 photographs per class, for a total of 180 images. The images were acquired under fixed viewpoint, background, and lighting conditions and were not used for model training or parameter selection. The model correctly classified all 180 images and mapped every

Table 7: Class-wise results of the controlled visual test.

Object	Images	Mean confidence	Mean time (ms)
Apple	30	0.771	49.89
Banana	30	0.948	48.02
Bottle	30	0.726	48.57
Paper cup	30	0.820	46.62
Mouse	30	0.914	46.79
Scissors	30	0.938	49.27

Note: All 180 images were classified correctly and mapped to the intended strategy.

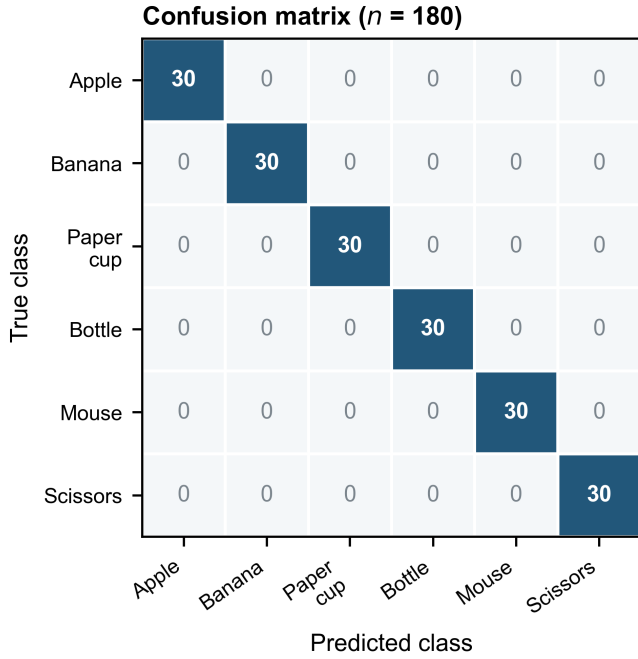


Figure 4: Controlled visual classification results for the separate 180-image test (30 images per class). Each cell reports the number of images with the indicated true and predicted class. All observations lie on the diagonal; every prediction mapped to the intended strategy.

prediction to the intended strategy. Mean confidence was 0.853, and mean wall-clock processing time was 48.19 ms per image (Figs. 4 and 5). These results characterize the six object instances under controlled imaging conditions but do not provide trial-level validation of online trigger behavior during teleoperation.

In the supervisory-loop implementation, the independent YOLO process, bounded queues, and non-blocking polling kept visual inference outside the synchronous call path of the nominal 200-Hz update. The median logged cycle time was approximately 5.07 ms, although the distribution had an upper tail with occasional large values.

4.2. Results Across the Five Modes

Mode C had the lowest median completion time, the highest observed success rate, and the lowest Raw NASA-TLX score

(Figs. 6 and 7). It also had the lowest median master-side trajectory length (Table 8), an exploratory measure. Based on the mean values, its completion time was 10.0%, 8.2%, 7.8%, and 8.5% lower than those of modes A, B, D, and E, respectively. Subsequent inferential analyses focused on completion time as the primary endpoint.

Consistent with the descriptive results, the Friedman test across the 27 matched task blocks indicated differences in completion time among the five modes ($\chi^2(4) = 30.904$, $p < 0.001$). Holm-adjusted paired Wilcoxon tests showed lower completion times in mode C than in modes A, B, D, and E (all $p < 0.01$, $r > 0.7$). Because the task blocks were nested within only three operators, these statistics describe repeated-measures differences within the present sample and do not support population-level inference across operators. Mode C also yielded the lowest Raw NASA-TLX score in each of the nine operator $\times$ strategy units, but three operators are insufficient for a population-level workload conclusion.

4.3. Primary Comparison: Full Cross-Channel Versus Impedance-Only Scheduling

The median completion times were 19.57 s in mode C and 20.73 s in mode E (Fig. 8). With $\Delta T = T_E - T_C$, the mean paired difference was 1.79 s and the bootstrap 95% CI was [1.10, 2.51] s, corresponding to a mean reduction of 8.5%. The mean differences for P01, P02, and P03 were 1.66, 2.56, and 1.16 s, respectively. Across the six objects, the relative differences ranged from 3.3% to 13.2% (Fig. 9).

The median master-side trajectory lengths were 0.697 m in mode C and 0.732 m in mode E. The mean paired difference was 0.024 m, with a bootstrap 95% CI of [-0.014, 0.059] m. Pause counts were lower on average in mode C but had the same median in modes C and E.

4.4. Exploratory Process Measure: C-E Pause Analysis

Pause count was analyzed as an exploratory process measure using master-side trajectories sampled at approximately 200 Hz. A pause was defined as a continuous interval in which master-device speed remained below 0.005 m/s for at least 0.30 s, excluding momentary decelerations and capturing more sustained interruptions. Mode C had a median pause count of 3 [2, 3.5] and a mean of 2.74 ± 1.23 , whereas mode E had a median of 3 and a mean of 3.41 ± 1.67 . Although mode C had a lower mean in each strategy class, the two modes had the same overall median. No formal hypothesis test was prespecified or performed for pause count, and no threshold-sensitivity analysis was available.

4.5. Failure Analysis

Eighteen failures occurred across the 135 trials: 5, 6, 1, 3, and 3 in modes A–E, respectively. The failures included object drops, observable slip, and visible damage, as summarized below. One designated researcher recorded outcomes in real time using the predefined criteria in Section 3.4; no systematic video recording or independent re-rating was available.

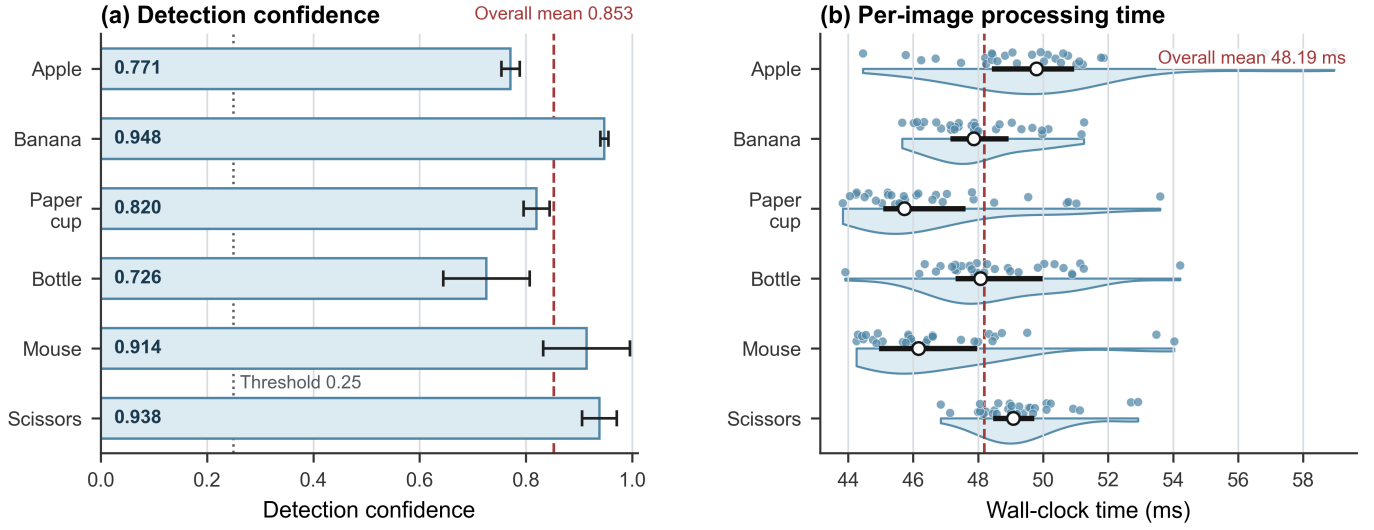


Figure 5: Confidence and processing time for the same controlled 180-image test. (a) Class-wise detection confidence: horizontal bars show mean $\pm$ SD; the dotted and dashed vertical lines mark the decision threshold (0.25) and overall mean (0.853). (b) Wall-clock processing time per image: half violins show class-wise distributions, dots show individual images, horizontal segments and open circles show the IQR and median, and the dashed vertical line marks the overall mean (48.19 ms).

Table 8: Results across the five experimental modes.

Mode	Completion time (s)	Trajectory length (m)	Success rate	Raw NASA-TLX
A	21.18 [20.62, 22.08]	0.757 [0.693, 0.816]	22/27 (81.5%)	62.50 [59.67, 64.50]
B	20.89 [20.12, 21.83]	0.787 [0.721, 0.861]	21/27 (77.8%)	56.17 [55.00, 59.33]
C	19.57 [18.41, 20.05]	0.697 [0.660, 0.769]	26/27 (96.3%)	48.67 [47.67, 51.83]
D	20.79 [20.32, 21.16]	0.722 [0.678, 0.768]	24/27 (88.9%)	60.33 [57.33, 62.50]
E	20.73 [19.95, 22.25]	0.732 [0.678, 0.799]	24/27 (88.9%)	53.67 [51.83, 57.83]

Note: Values are median [IQR], except success rate. Mode definitions are given in Table 3; trial-level data are provided in Supplementary Table S1.

Table 9: Primary comparison of full cross-channel and impedance-only scheduling.

Measure	C, median [IQR]	E, median [IQR]	Mean Δ (E–C), 95% CI	Operator direction
Completion time (s)	19.57 [18.41, 20.05]	20.73 [19.95, 22.25]	1.79 [1.10, 2.51]	3/3 favored C
Trajectory length (m)	0.697 [0.660, 0.769]	0.732 [0.678, 0.799]	0.024 [-0.014, 0.059]	Mixed
Raw NASA-TLX	48.67 [47.67, 51.83]	53.67 [51.83, 57.83]	4.87 [not calculated]	3/3 favored C

Note: Positive Δ values favor mode C. Intervals are matched-task-block bootstrap 95% CIs from 10,000 resamples; no bootstrap interval was calculated for Raw NASA-TLX.

Table 10: Observed failures and representative observations by experimental mode.

Mode	Failures/total	Representative observations
A	5/27	Paper-cup deformation and unstable positioning of the scissors.
B	6/27	The manual-selection workflow included additional judgment and switching steps; the available records do not support attribution of specific failure causes.
C	1/27	Slip of the mouse during transfer due to its smooth surface.
D	3/27	Object drops or loss of grasp during holding.
E	3/27	Instability during grasp establishment or transfer for some objects.

Note: Mode definitions are given in Table 3. Observations are descriptive and were recorded in real time.

Mode C had one failure in 27 trials, the lowest observed conditions. count among the five modes under the present experimental

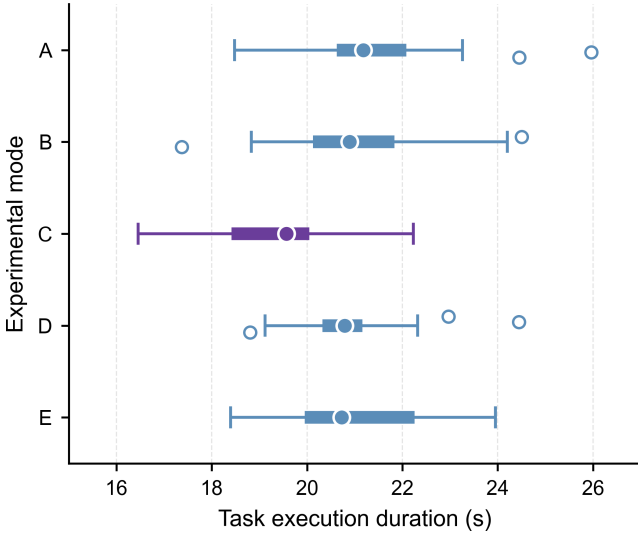


Figure 6: Task execution duration across the five modes: fixed parameters (A), operator selection (B), full cross-channel scheduling (C), visual-semantic cue only (D), and impedance-only scheduling (E). Each mode contains 27 matched task blocks. Filled circles show medians, thick horizontal segments show IQRs, and thin horizontal segments with end caps show $1.5 \times$ IQR Tukey whiskers; small open circles show observations beyond the whiskers. Mode C was lower than modes A, B, D, and E in Holm-adjusted paired Wilcoxon tests (all $p < 0.01$).

Table 11: Mean completion time by object and experimental mode.

Object	A	B	C	D	E
Apple	20.46	21.40	19.25	20.81	20.94
Banana	20.07	20.85	19.49	21.06	20.64
Paper cup	22.36	20.27	18.69	20.38	21.17
Bottle	22.11	22.14	19.63	20.78	20.30
Mouse	21.74	20.93	19.75	20.74	21.64
Scissors	21.79	20.70	18.84	21.83	21.70

Note: Values are mean completion time in seconds. Mode definitions are given in Table 3; bold indicates the lowest mean in each row.

4.6. Consistency Across Operators and Objects

Figure 9 reports paired C–E differences by operator and object. At the operator level, the mean paired differences $\Delta T = T_E - T_C$ for P01, P02, and P03 were 1.66, 2.56, and 1.16 s, respectively. The corresponding median [IQR] differences were 1.91 [0.60, 3.14], 2.63 [2.19, 3.34], and 1.60 [−0.53, 1.73] s. Mode C was faster in 7/9, 9/9, and 6/9 task blocks, respectively. After P01, P02, and P03 were excluded in turn, the mean differences in the remaining blocks were 1.86, 1.41, and 2.11 s, respectively; all continued to favor mode C.

At the object level, mode C also had the shortest mean completion time among the five modes for each of the six objects:

Relative to mode E, the mean completion-time reductions in mode C were 3.3% for the bottle, 5.5% for the banana, 8.1% for the apple, 8.7% for the mouse, 11.7% for the paper cup, and 13.2% for the scissors. The direction was consistent across all objects, although the magnitude varied.

5. Discussion

5.1. Interpreting the Five-Mode Evaluation

1. Semantic display and automatic configuration play distinct experimental roles. The five modes differed in how they used object information and which channels they reconfigured. Mode A used a fixed parameter vector without a visual interface. Mode B also omitted the visual interface; after timing began, the operator identified the object, selected a strategy, and only then initiated robot motion. The B–C difference therefore includes removal of a manual decision-and-selection step and does not isolate controller performance. Mode D displayed the same class, mapped strategy, confidence, and bounding box as mode C but retained fixed parameters. The C–D contrast is therefore the clearest test of automatic full-parameter reconfiguration beyond the visual-semantic cue itself.

Mode C was faster and yielded lower Raw NASA-TLX scores than mode D while also reconfiguring the full parameter vector. This comparison indicates an additional task-level benefit of automatic parameter configuration beyond the default visual-semantic display. By contrast, the D–A comparison represents the effect of the complete displayed cue rather than that of the class label alone.

Overall, mode C had the shortest completion time and the lowest Raw NASA-TLX score in the present sample. The C–D and C–E comparisons indicate a task-level benefit of automatic parameter reconfiguration that was not reproduced by either the default visual-semantic display or impedance-only reconfiguration. These conclusions apply to the designed default interfaces. The C–E comparison provides the primary system-level evidence for the complete parameter bundle.

5.2. System-Level Effect of the Complete Parameter Bundle

2. Impedance reconfiguration alone did not reproduce the performance of the full bundle. The C–E comparison is the primary system-level contrast in this study. Both modes used the same class-to-strategy mapping, impedance settings, and default visual-semantic interface. Mode C additionally updated the master-side haptic gain, force dead zone, gripper speed, and requested grasp-force setting. Because the comparison removes the haptic-interface and gripper parameter groups together, it assesses the overall difference between the complete parameter bundle and impedance-only scheduling but cannot identify the contribution of each added channel.

Empirically, the paired mean E–C completion-time difference was 1.79 s, with a matched-block bootstrap 95% CI of [1.10, 2.51] s. The confidence interval for master-side trajectory length crossed zero, indicating no clear reduction in geometric path length. Mode C also had a lower mean pause count, but the records do not localize this difference to grasp, transfer, or release and contain no separate measure of corrective action.

The shorter completion time in mode C is therefore associated with the added haptic-interface and gripper settings as a bundle. Impedance settings determine the robot response to pose error and disturbance, master-side settings alter how estimated external force is rendered, and gripper settings govern the

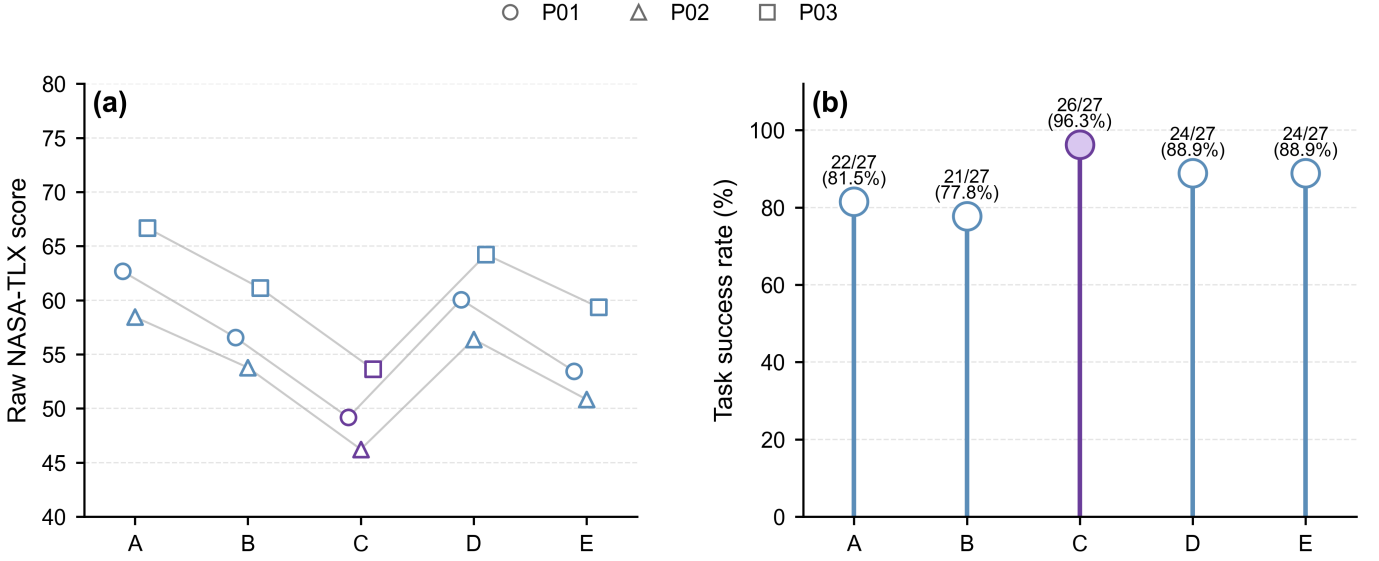


Figure 7: Human workload and observed task success across the five modes. (a) Raw NASA-TLX: connected markers show each operator’s mean across three questionnaire units per mode; marker shape identifies the operator. (b) Observed successful trials out of 27. Mode C had the lowest Raw NASA-TLX score and the highest observed success rate; these results are descriptive within the tested sample.

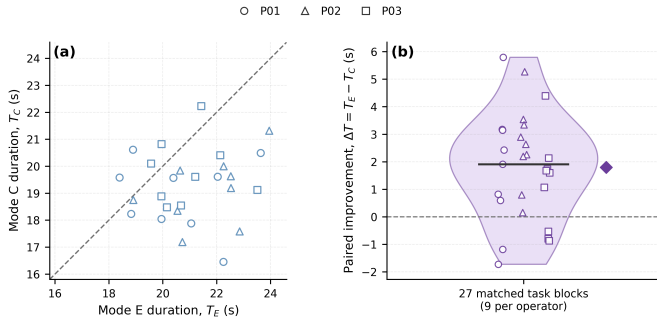


Figure 8: Paired completion-time comparison between modes C and E (27 matched blocks across three operators). (a) Points below the identity line indicate a shorter duration in mode C. (b) Paired improvement, $\Delta T = T_E - T_C$; positive values favor mode C. The violin, horizontal segment, and diamond show the distribution, median, and mean improvement (1.79 s; bootstrap 95% CI [1.10, 2.51] s), respectively.

speed and requested force used to establish the grasp. The experiment evaluates the latter two parameter groups together; it neither identifies their separate contributions nor demonstrates that the commanded Franka Hand force equals the actual fingertip contact force. Channel-specific ablations are required before the observed difference can be attributed to either group separately.

The trajectory-length interval crossed zero. Thus, the completion-time advantage of mode C cannot be attributed simply to a shorter geometric path. The pause-count result remains exploratory and does not establish fewer interruptions, greater movement continuity, or any causal mechanism, particularly because no separate correction or regrasp measure was available.

Because the configuration was evaluated as a bundle, the

failure results likewise cannot be attributed to any single parameter channel.

5.3. Asynchronous Perception–Control Integration

3. A one-shot update does not require perception to operate at the supervisory-loop rate. Image acquisition at 15 fps and independent inference exchanged data with the nominal 200-Hz update through bounded queues. The single-slot frame queue limited stale images, while non-blocking polling prevented the supervisory loop from waiting for visual output. This architecture is appropriate when perception supports a single configuration event after task initiation rather than continuous visual servoing.

The implementation therefore demonstrates asynchronous, non-blocking integration, but it does not establish hard real-time performance.

5.4. Variation Across Operators and Objects

At the operator level, the mean paired difference was positive for all three operators and remained positive when each operator was excluded in turn. Nevertheless, the participant sample was limited to three individuals.

At the object level, mode C was faster than mode E for all six objects, with mean reductions ranging from 3.3% to 13.2%. The differences were smaller for the bottle and banana and larger for the paper cup and scissors. This variation may reflect differences in grasp pose, deformation risk, and orientation-stability requirements. However, because only four or five matched task blocks were available for each object, the direction and magnitude are reported descriptively without assigning a specific mechanism to the object-level differences.

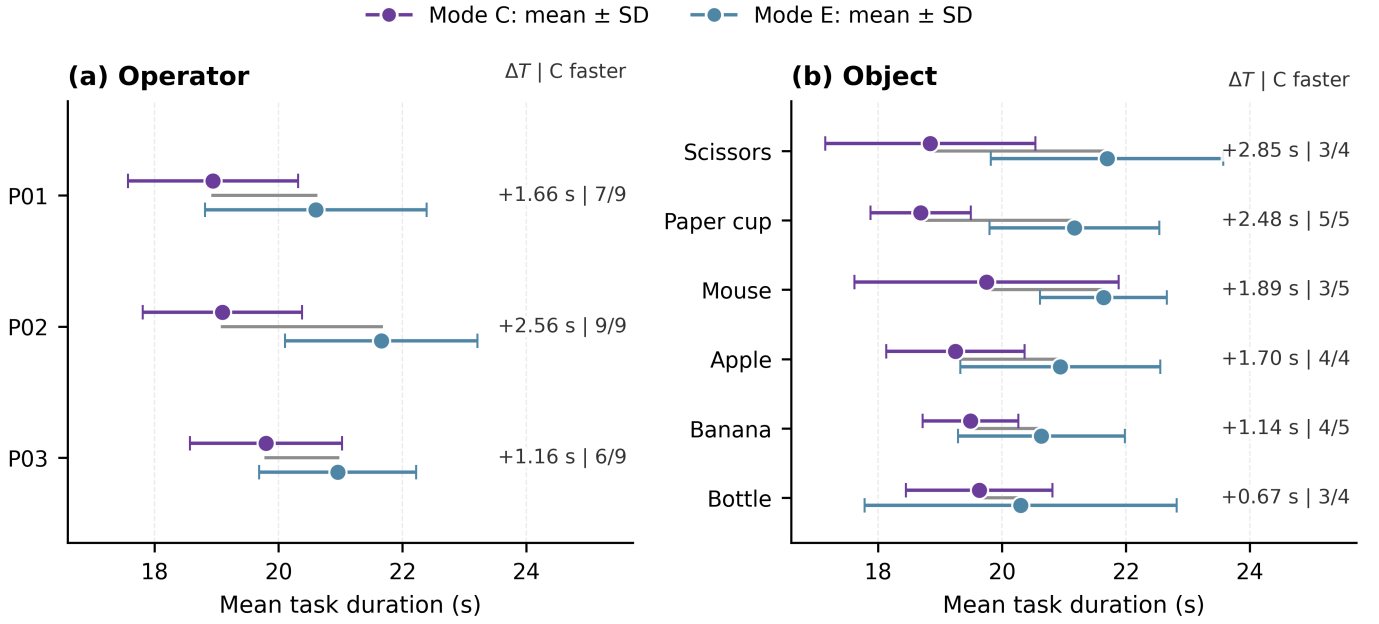


Figure 9: Paired C-E mean task duration by operator and object. Purple and blue points show modes C and E (mean $\pm$ SD), and gray connectors link the paired means. In every displayed stratum, mode C has the lower mean duration. Labels give $\Delta T = T_E - T_C$ and the number of matched blocks favoring C (9 per operator; 4–5 per object). Results are descriptive.

5.5. Relation to Previous Work and Mechatronic Design Implications

Previous tele-impedance and variable-impedance approaches commonly adapt remote impedance according to human state, contact force, demonstrations, or task phase [6–9, 18–21]. Visual methods have translated object attributes, geometry, and vision–language inputs into stiffness or impedance matrices [10, 11, 13, 14], while automatic switching, shared control, and multi-stiffness interfaces have addressed mode selection and control-authority allocation [12, 15, 16, 23, 24]. The present work does not claim the first use of vision to adjust impedance. Its distinction lies in its system-level formulation and evaluation design: an operational strategy assigns a coordinated parameter bundle to the slave-side mechanical response, the operator-facing haptic interface, and gripper execution. This coordination is semantic and task-level rather than a dynamically derived coupling law. Bounded queues decouple low-rate perception from the nominal high-rate supervisory update, while strategy locking maintains a single configuration within each trial. The five-mode experiment then compares the complete configuration with fixed parameters, manual selection, a visual-semantic cue with fixed parameters, and impedance-only scheduling.

From a design perspective, this principle may be useful in flexible sorting, remote maintenance, and unstructured dismantling, where a limited set of recurring object or task classes can be associated with validated operating strategies. Scaling beyond a small class set would require systematic maintenance of the mapping and potentially more than three strategy levels. Additional sensing, such as tactile or contact-state information, could support post-contact verification or correction without placing the visual process in the high-rate supervisory path.

5.6. Limitations and Future Work

A timing-verification limitation is that the formal trial logs did not retain complete visual-event histories or synchronized strategy-event–contact timestamps. The trigger–contact interval could therefore not be reconstructed retrospectively for individual trials. This limitation affects quantitative verification of the pre-contact timing margin, but it does not alter the task-level C–E comparison of the complete parameter bundle. The 180-image test characterizes recognition under controlled imaging conditions but does not provide trial-level evidence of trigger correctness, fallback use, interface visibility, or completion before contact during the 135 teleoperation trials.

The study involved only three operators, with repeated measurements from the same participants. The inferential tests therefore describe within-sample repeated-measures differences; a larger participant study is required to assess population-level generalizability. All operators were right-handed but operated the left-positioned Omega.7 with the left hand. Only translational master motion was mapped, and the end-effector orientation remained fixed. Objects were manually reset within a prescribed region and approximate orientation, while the nine preassigned mode sequences did not provide complete positional balance. The results therefore apply to the present device layout, training conditions, motion mapping, object-placement protocol, and participant sample.

The class-to-strategy mapping is also manual. A new label can be added without controller retraining only if the detector already recognizes it reliably; otherwise, the detector must be replaced or retrained. The system does not automatically infer unseen physical properties or an appropriate strategy. The three strategy levels are practical abstractions for the present task and

may become insufficient as the number and diversity of objects increase.

The parameter sets were operating points selected through qualitative screening rather than optimization or material-strength testing. Because the C–E comparison removes the haptic-interface and gripper parameter groups together, it cannot separate their contributions. Future ablations should independently remove haptic-gain, dead-zone, and gripper scheduling. The baseline Omega.7 force-feedback command had no additional software saturation, and saturation status was not logged. Future implementations should add an explicit software force limit and record both commanded and limited outputs. The Franka Hand force value specified a requested grasp setting rather than a measured fingertip force; direct fingertip-force measurements are needed to validate its physical effect.

Success and failure were judged in real time by one designated researcher using predefined criteria. Trials were not systematically video-recorded or independently re-rated, so slip and visible-damage labels retain an observational component. Failure did not end timing immediately, and a dropped object could not be regrasped; instead, the operator completed the common terminal procedure. This protocol avoids artificially short failure times but does not provide stage-specific failure durations. The records also lack a separate selection time for mode B, quantitative contact-quality measures, and a direct measure of corrective action.

Pause count used an engineering threshold of 0.005 m/s sustained for at least 0.30 s. No threshold-sensitivity analysis or prespecified inferential test was performed, so this result remains exploratory. Visual inference was kept outside the synchronous call path of the nominal 200-Hz update, but the cycle-time logs contained occasional large values; hard real-time performance was therefore not established. Future work should add synchronized perception, contact, interface, and mode-selection logs; perform pause-threshold sensitivity analysis; record video for independent outcome assessment; incorporate active orientation control and less constrained object placement; and recruit a larger, more diverse participant sample.

6. Conclusion

This paper presents a vision-semantic, event-triggered multi-channel joint parameter-configuration framework for pre-contact parameter preparation in haptic teleoperation of heterogeneous objects. A three-level mapping from object semantics to an operational strategy and then to a joint parameter set aims to make slave-side impedance, the master-side haptic interface, and gripper execution available as a unified strategy-level bundle before interaction. An asynchronous one-shot update integrates this configuration after task initiation while keeping low-rate perception outside the synchronous supervisory path. By avoiding complex online optimization, the framework provides interpretable and deployable system-level parameter configuration with a relatively low online computational burden.

In 135 physical trials involving three operators, six objects, and five modes, the full cross-channel mode C achieved the shortest completion time, the highest observed success rate, and

the lowest Raw NASA-TLX score. Relative to mode E, which used the same semantic mapping and impedance settings, mode C reduced mean completion time by 1.79 s, with a matched-task-block bootstrap 95% CI of [1.10, 2.51] s. The direction of the C–E difference favored mode C for each operator and object. No clear difference was observed in master-side trajectory length, indicating that the time advantage could not be explained solely by path shortening.

Under the present heterogeneous-object teleoperation conditions, the complete configuration produced better task-level performance than either impedance-only scheduling or the default visual-semantic cue with fixed parameters. The experiments provide system-level evidence for the overall effect of the complete parameter bundle on the current platform, while the causal contribution of each channel remains unresolved.

Looking ahead, this work will extend early-stage parameter preparation toward closed-loop multi-channel configuration throughout the manipulation process. First, synchronized records of visual strategy events, initial contact, external-force estimates, gripper states, and task outcomes will be added to quantify the temporal relationship between semantic triggering, contact establishment, and task performance. Second, post-contact feedback, including interaction force, slip risk, and grasp state, will be used to verify and, when needed, correct the parameter bundle selected by the strategy event. This will advance the system from one-shot preparation to a “pre-contact prediction–in-contact verification–post-contact adjustment” loop. Finally, strategy learning, parameter-sensitivity analysis, and operator-specific adaptation will be studied across broader object classes and task conditions to improve generalizability and long-term deployability.

Supplementary Material

Supplementary Tables S1 and S2 are supplied as a separate editable supplementary file.

Declarations

Ethics statement

SUBMISSION-BLOCKING PLACEHOLDER: This study involved three adult volunteers. *Before submission, replace this sentence with the written determination from the relevant institutional committee or office, including the institution, decision type (approval, exemption, or no formal review required), date, reference number if available, and the actual basis for that determination.* All participants were informed of the experimental procedures and provided written informed consent before participation. No personally identifiable information was collected; all data were anonymized and used solely for academic research.

Image integrity

Figure 1 is a photograph of the actual experimental platform. No generative AI was used to add, remove, replace, or reposition experimental elements. Figures 2–3 were redrawn by

the authors using Python/Matplotlib, and Figures 4–9 were generated from experimental data. The authors verified the logic, text, and data mapping in the final figures.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

De-identified trial data are available from the corresponding author upon reasonable request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex for language editing, consistency checks, early layout discussions, and assistance with drafting Python/Matplotlib code. The authors reviewed and edited the content as needed, verified all technical statements and data mappings, and take full responsibility for the content of the manuscript.

References

- [1] D. Lawrence, Stability and transparency in bilateral teleoperation, *IEEE Trans Robot Autom* 9 (5) (1993) 624–37. doi:10.1109/70.258054.
- [2] C. Passenberg, A. Peer, M. Buss, A survey of environment-, operator-, and task-adapted controllers for teleoperation systems, *Mechatronics* 20 (7) (2010) 787–801. doi:10.1016/j.mechatronics.2010.04.005.
- [3] N. Hogan, Impedance control: An approach to manipulation: Part I—theory, *J Dyn Syst Meas Control* 107 (1) (1985) 1–7. doi:10.1115/1.3140702.
- [4] K. Kronander, A. Billard, Stability considerations for variable impedance control, *IEEE Trans Robot* 32 (5) (2016) 1298–305. doi:10.1109/TRO.2016.2593492.
- [5] F. Abu-Dakka, M. Saveriano, Variable impedance control and learning—a review, *Front Robot AI* 7 (2020) 590681. doi:10.3389/frobt.2020.590681.
- [6] D. Walker, R. Wilson, G. Niemeyer, User-controlled variable impedance teleoperation, in: *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2010, pp. 5352–57. doi:10.1109/ROBOT.2010.5509811.
- [7] A. Ajoudani, N. Tsagarakis, A. Bicchi, Tele-impedance: Teleoperation with impedance regulation using a body–machine interface, *Int J Robot Res* 31 (13) (2012) 1642–56. doi:10.1177/0278364912464668.
- [8] M. Laghi, A. Ajoudani, M. Catalano, A. Bicchi, Unifying bilateral teleoperation and tele-impedance for enhanced user experience, *Int J Robot Res* 39 (4) (2020) 514–39. doi:10.1177/0278364919891773.
- [9] Y. Michel, R. Rahal, C. Pacchierotti, P. R. Giordano, D. Lee, Bilateral teleoperation with adaptive impedance control for contact tasks, *IEEE Robot Autom Lett* 6 (3) (2021) 5429–36. doi:10.1109/LRA.2021.3066974.
- [10] Y.-C. Huang, D. Abbink, L. Peternel, A semi-autonomous tele-impedance method based on vision and voice interfaces, in: *Proceedings of the 20th International Conference on Advanced Robotics (ICAR)*, 2021, pp. 180–6. doi:10.1109/ICAR53236.2021.9659427.
- [11] A. Oliva, P. Giordano, F. Chaumette, A general visual-impedance framework for effectively combining vision and force sensing in feature space, *IEEE Robot Autom Lett* 6 (3) (2021) 4441–8. doi:10.1109/LRA.2021.3068911.
- [12] M. Bowman, J. Zhang, X. Zhang, Intent-based task-oriented shared control for intuitive telemanipulation, *J Intell Robot Syst* 110 (4) (2024) 167. doi:10.1007/s10846-024-02185-1.
- [13] G. Siegemund, A. Díaz Rosales, A. Glodde, F. Dietrich, L. Peternel, Semi-autonomous teleimpedance based on visual detection of object geometry and material and its relation to environment, in: *Proceedings of the IEEE-RAS 23rd International Conference on Humanoid Robots (Humanoids)*, 2024, pp. 779–86. doi:10.1109/Humanoids58906.2024.10769858.
- [14] H. H. A. Jekel, A. Díaz Rosales, L. Peternel, Visio-verbal teleimpedance interface: Enabling semi-autonomous control of physical interaction via eye tracking and speech, *Front Robot AI* 13 (2026) 1749105. doi:10.3389/frobt.2026.1749105.
- [15] W. Li, F. Huang, Z. Chen, Z. Chen, Automatic-switching-based teleoperation framework for mobile manipulator with asymmetrical mapping and force feedback, *Mechatronics* 99 (2024) 103164. doi:10.1016/j.mechatronics.2024.103164.
- [16] R. Balachandran, M. D. Stefano, H. Mishra, C. Ott, A. Albu-Schäffer, Passive arbitration in adaptive shared control of robots with variable force and stiffness scaling, *Mechatronics* 90 (2023) 102930. doi:10.1016/j.mechatronics.2022.102930.
- [17] S. Park, Y. Park, J. Bae, Performance evaluation of a tactile and kinesthetic finger feedback system

- for teleoperation, *Mechatronics* 87 (2022) 102898. doi:10.1016/j.mechatronics.2022.102898.
- [18] R. Li, M. Cheng, R. Ding, Passivity-based bilateral shared variable impedance control for teleoperation compliant assembly, *Mechatronics* 95 (2023) 103057. doi:10.1016/j.mechatronics.2023.103057.
- [19] Z. Wang, X. Xu, D. Yang, B. Güleçyüz, F. Meng, E. Steinbach, Teleoperation with haptic sensor-aided variable impedance control based on environment and human stiffness estimation, *IEEE Sens J* 24 (14) (2024) 22168–77. doi:10.1109/JSEN.2024.3369758.
- [20] Y. Michel, Y. Abdelhalem, G. Cheng, Passivity-based teleoperation with variable rotational impedance control, *IEEE Robot Autom Lett* 9 (12) (2024) 11658–65. doi:10.1109/LRA.2024.3490260.
- [21] H. Lee, J. Han, G.-H. Yang, Development of variable scaling teleoperation framework for improving teleoperation performance, *Int J Control Autom Syst* 22 (3) (2024) 936–45. doi:10.1007/s12555-022-1099-z.
- [22] M. Lippi, M. Welle, M. Wozniak, A. Gasparri, D. Kragic, Low-cost teleoperation with haptic feedback through vision-based tactile sensors for rigid and soft object manipulation, in: *Proceedings of the 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, 2024, pp. 1963–9. doi:10.1109/RO-MAN60168.2024.10731383.
- [23] A. Díaz Rosales, J. Rodriguez-Nogueira, E. Matheson, D. Abbink, L. Peternel, Interactive multi-stiffness mixed reality interface: Controlling and visualizing robot and environment stiffness, in: *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 13479–86. doi:10.1109/IROS58592.2024.10801866.
- [24] B. Güleçyüz, R. Balachandran, M. Panzirsch, H. Singh, T. Hulin, X. Xu, et al., Enhancing shared autonomy in teleoperation under network delay: Transparency- and confidence-aware arbitration, *IEEE Robot Autom Lett* 10 (10) (2025) 9654–61. doi:10.1109/LRA.2025.3596436.
- [25] N. Riazat, O. Erin, A. Krieger, J. Brown, Investigating haptic feedback in vision-deficient millirobot telemanipulation, *IEEE Robot Autom Lett* 9 (7) (2024) 6178–85. doi:10.1109/LRA.2024.3397529.
- [26] S. Haddadin, S. Parusel, L. Johannsmeier, S. Golz, S. Gabl, F. Walch, et al., The Franka Emika robot: A reference platform for robotics research and education, *IEEE Robot Autom Mag* 29 (2) (2022) 46–64. doi:10.1109/MRA.2021.3138382.
- [27] S. Hart, NASA task load index (NASA-TLX); 20 years later, in: *Proc Hum Factors Ergon Soc Annu Meet*, Vol. 50, 2006, pp. 904–8. doi:10.1177/154193120605000909.